Stat 526 Hw 7

Bowen Zheng

2026-03-24

Problem 0

Bowen Zheng

Problem 1: Faraway Chapter 7 Exercise #2

> Data were collected from 39 students in a University of Chicago MBA class and may be found in the dataset happy.

a.)

> Make plots of the relationship between the response, happiness and each of the four predictors. Transform sex and love to appropriate factors.

We can see what the data means here:

https://rdrr.io/cran/faraway/man/happy.html

```
library(faraway)
data(happy)
happy$sex <- factor(happy$sex, labels = c("Unsatisfactory", "Satisfactory"))
happy$love <- factor(happy$love, labels = c("Lonely", "Secure", "Belonging"))
print(happy)
```

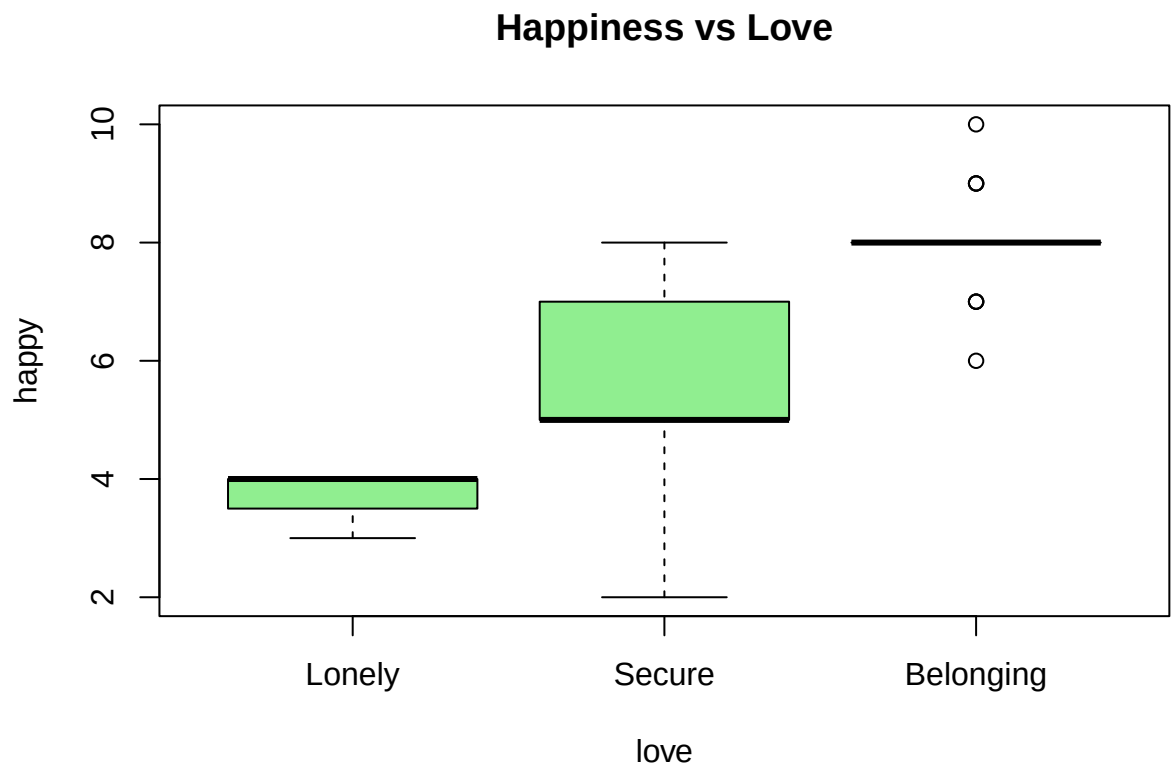
```
##    happy money            sex      love work
## 1     10    36 Unsatisfactory Belonging    4
## 2      8    47   Satisfactory Belonging    1
## 3      8    53 Unsatisfactory Belonging    5
## 4      8    35   Satisfactory Belonging    3
## 5      4    88   Satisfactory    Lonely    2
## 6      9   175   Satisfactory Belonging    4
## 7      8   175   Satisfactory Belonging    4
## 8      6    45 Unsatisfactory    Secure    3
## 9      5    35   Satisfactory    Secure    2
## 10     4    55   Satisfactory    Lonely    4
## 11     7    40 Unsatisfactory    Secure    3
## 12     8    45   Satisfactory Belonging    4
## 13     8    45   Satisfactory Belonging    3
```

```
## 14      8    45   Satisfactory Belonging    4
## 15      8    62 Unsatisfactory Belonging    4
## 16      4    44   Satisfactory    Secure    3
## 17      8    85   Satisfactory Belonging    4
## 18      8    81 Unsatisfactory    Secure    4
## 19      6   112   Satisfactory Belonging    2
## 20      5    40   Satisfactory    Secure    4
## 21      7    40   Satisfactory Belonging    3
## 22      9    44   Satisfactory Belonging    4
## 23      4    35   Satisfactory    Secure    2
## 24      5    56   Satisfactory    Secure    3
## 25      9   115   Satisfactory Belonging    4
## 26      7    44   Satisfactory    Secure    4
## 27      7    50   Satisfactory Belonging    3
## 28      7    45 Unsatisfactory    Secure    4
## 29      8    41 Unsatisfactory Belonging    5
## 30      8    50 Unsatisfactory Belonging    5
## 31      3    85   Satisfactory    Lonely    1
## 32      7    90   Satisfactory    Secure    2
## 33      7    85   Satisfactory    Secure    4
## 34      8    75   Satisfactory Belonging    4
## 35      5    70 Unsatisfactory    Secure    3
## 36      2     0 Unsatisfactory    Secure    2
## 37      5    31 Unsatisfactory    Secure    4
## 38      7    60   Satisfactory Belonging    4
## 39      8    65   Satisfactory Belonging    3
```

```r
plot(happy ~ money, data=happy, main="Happiness vs Money", pch=19)
```

```r
plot(happy ~ sex, data=happy, main="Happiness vs Sex", col="lightblue")
```

```r
plot(happy ~ love, data=happy, main="Happiness vs Love", col="lightgreen")
```




```r
plot(happy ~ work, data=happy, main="Happiness vs Work", pch=19)
```

b.)

> Fit a proportional odds model with the main effects of each of the four predictors. Extract the estimated values of using the zeta component of the fit. Make an index plot of these and comment on any deviation from linearity and what this may signify.

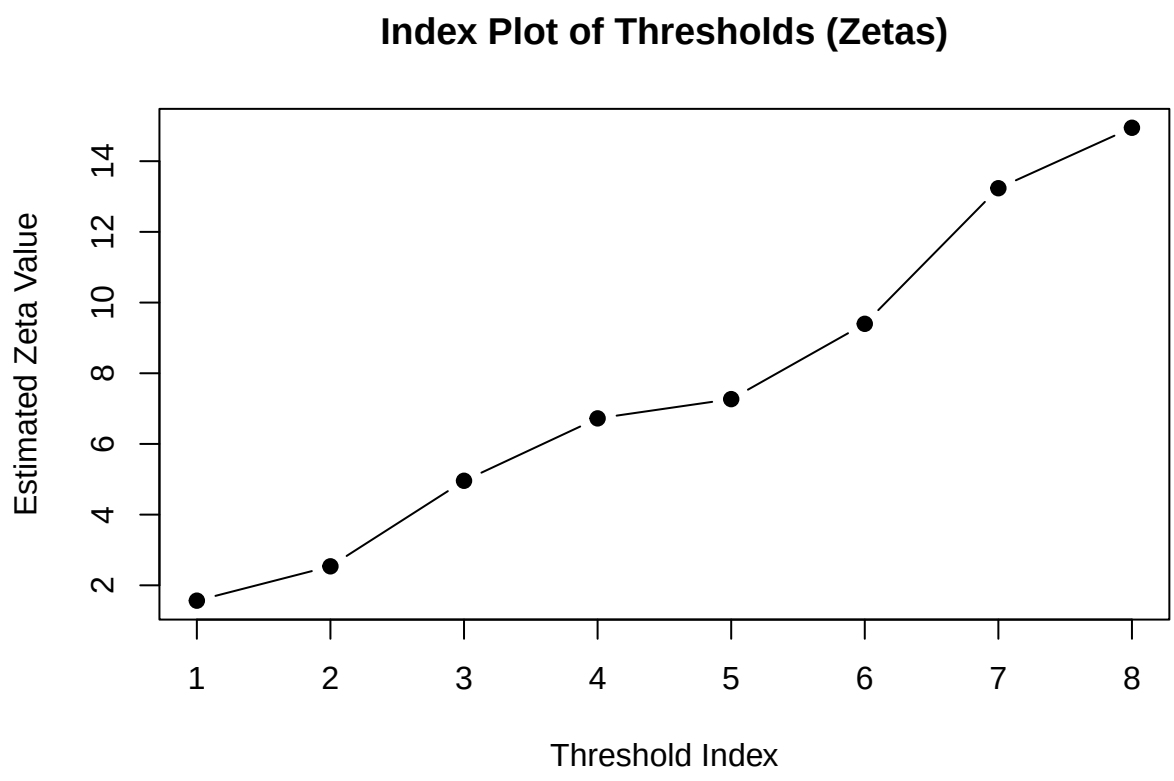
We use polr from library MASS like Prof Song.

```r
library(MASS)

model_prop_odds <- polr(factor(happy) ~ money + sex + love + work, data = happy)
summary(model_prop_odds)
```

```
##
## Re-fitting to get Hessian

## Call:
## polr(formula = factor(happy) ~ money + sex + love + work, data = happy)
##
## Coefficients:
##                    Value Std. Error t value
## money            0.02195    0.01073  2.0456
## sexSatisfactory -0.57364    0.81896 -0.7005
## loveSecure       2.95768    1.40285  2.1083
```

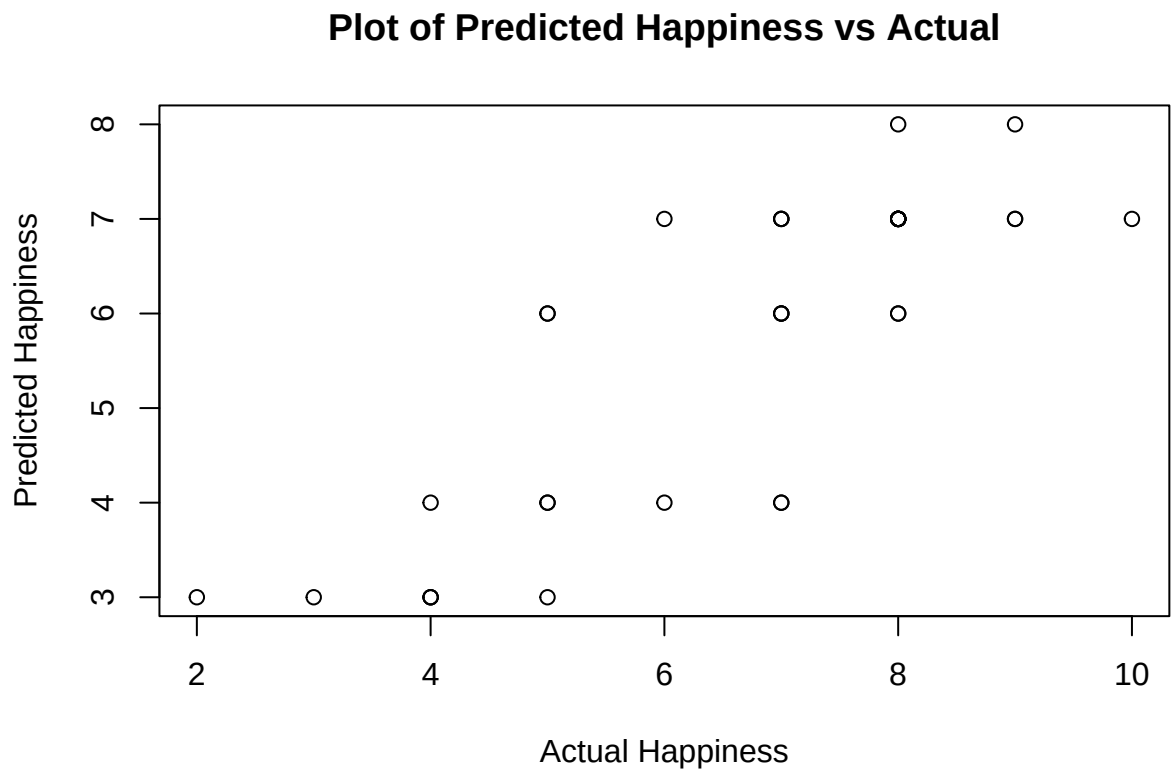
```
## loveBelonging    6.89136    1.67259  4.1202
## work             0.88821    0.40764  2.1789
##
## Intercepts:
##      Value   Std. Error t value
## 2|3   1.5654  1.8376     0.8519
## 3|4   2.5373  1.7712     1.4325
## 4|5   4.9572  2.0410     2.4289
## 5|6   6.7230  2.1690     3.0995
## 6|7   7.2680  2.1979     3.3067
## 7|8   9.3987  2.3380     4.0200
## 8|9  13.2350  2.7378     4.8341
## 9|10 14.9454  2.9396     5.0842
##
## Residual Deviance: 94.57552
## AIC: 120.5755
```

```r
plot(model_prop_odds$zeta, type="b", pch=19, xlab="Threshold Index", ylab="Estimated Zeta Value", main=
```




```r
predicted_happiness <- predict(model_prop_odds, type = "class")
plot(predicted_happiness, pch=19, xlab="Threshold Index", ylab="Count", main="Plot of Predicted Happines
```

```
plot(happy$happy, predicted_happiness, ylab="Predicted Happiness", xlab="Actual Happiness", main="Plot o
```



Does not seem very linear. This is probably due to most of the predicted values being 4,5,7,8,9. The actual model has much more varied responses.

c.)

> Use stepwise AIC to select a reduced model. Which variables are eliminated. Make a test to compare this reduced model to the original model.

Stepwise AIC first:

```
reduced_model <- step(model_prop_odds, direction = "both", trace = 0)
summary(reduced_model)
```

```
##
## Re-fitting to get Hessian

## Call:
## polr(formula = factor(happy) ~ money + love + work, data = happy)
##
## Coefficients:
##                  Value Std. Error t value
## money          0.01996    0.01022   1.953
## loveSecure     3.05415    1.40034   2.181
## loveBelonging 6.78758    1.65935   4.090
```

```
## work          0.97929    0.38805   2.524
##
## Intercepts:
##      Value   Std. Error t value
## 2|3   2.1434  1.6458     1.3023
## 3|4   3.1493  1.5522     2.0290
## 4|5   5.6216  1.8313     3.0697
## 5|6   7.3229  2.0085     3.6460
## 6|7   7.8633  2.0446     3.8459
## 7|8   9.9858  2.2045     4.5298
## 8|9  13.7281  2.6402     5.1997
## 9|10 15.4434  2.8495     5.4197
##
## Residual Deviance: 95.07054
## AIC: 119.0705
```

Anova test comparing nested models deviance next:

```r
anova(model_prop_odds, reduced_model)
```

```
## Likelihood ratio tests of ordinal regression models
##
## Response: factor(happy)
##                        Model Resid. df Resid. Dev   Test    Df  LR stat.
## 1       money + love + work        27   95.07054
## 2 money + sex + love + work        26   94.57552 1 vs 2     1 0.4950246
##     Pr(Chi)
## 1
## 2 0.4816945
```

We removed sex but it does not seem like removing sex was necessary as the p-value of the comparison is quite high. It is not significantly better to remove it.

d.)

> Compare a model where love is treated as an ordered factor with one where it is treated as a numerical predictor. Which model is preferred?

```r
# Model with love as Numerical
data(happy)
happy_numeric <- happy
happy_numeric$sex <- factor(happy_numeric$sex, labels = c("Unsatisfactory", "Satisfactory"))
model_numeric <- polr(factor(happy) ~ money + sex + love + work, data = happy_numeric)

anova(model_numeric, model_prop_odds)
```

```
## Likelihood ratio tests of ordinal regression models
##
## Response: factor(happy)
##                        Model Resid. df Resid. Dev   Test    Df  LR stat.
## 1 money + sex + love + work        27   94.86029
```

```
## 2 money + sex + love + work        26   94.57552 1 vs 2     1 0.2847774
##     Pr(Chi)
## 1
## 2 0.5935869
```

```r
AIC(model_numeric, model_prop_odds)
```

```
##                 df      AIC
## model_numeric   12 118.8603
## model_prop_odds 13 120.5755
```

Using ANOVA (likelihood test) we see that the factor version has lower deviance, but not significantly so. It also has lower AIC. We prefer the original factor version.

e.)

> Examine the regression coefficients of your selected model and provide an interpretation of their meaning.

```r
happy$sex <- factor(happy$sex, labels = c("Unsatisfactory", "Satisfactory"))
happy$love <- factor(happy$love, labels = c("Lonely", "Secure", "Belonging"))
summary(model_prop_odds)
```

```
##
## Re-fitting to get Hessian

## Call:
## polr(formula = factor(happy) ~ money + sex + love + work, data = happy)
##
## Coefficients:
##                    Value Std. Error t value
## money            0.02195    0.01073  2.0456
## sexSatisfactory -0.57364    0.81896 -0.7005
## loveSecure       2.95768    1.40285  2.1083
## loveBelonging    6.89136    1.67259  4.1202
## work             0.88821    0.40764  2.1789
##
## Intercepts:
##      Value   Std. Error t value
## 2|3   1.5654  1.8376     0.8519
## 3|4   2.5373  1.7712     1.4325
## 4|5   4.9572  2.0410     2.4289
## 5|6   6.7230  2.1690     3.0995
## 6|7   7.2680  2.1979     3.3067
## 7|8   9.3987  2.3380     4.0200
## 8|9  13.2350  2.7378     4.8341
## 9|10 14.9454  2.9396     5.0842
##
## Residual Deviance: 94.57552
## AIC: 120.5755
```

```r
exp(coef(model_prop_odds))
```

```
##           money sexSatisfactory      loveSecure   loveBelonging            work
##       1.0221932       0.5634688      19.2533389     983.7402862       2.4307738
```

Our logit link proportional odds model sees positive coefficients as probability of going to a higher category increases for that predictor. We see that love lonely is the base starting point. So using the log odds conversion, money increases happiness the least. Next is work, then love secure, and the most is love Belonging. Having satisfactory sex seems to decrease happiness.

f.)

> Varying money across its range and holding love and work fixed at their median values, predict the most likely happiness level. Repeat the process varying only love and then work. What do we learn about the effect of the three predictors on the response?

Let us just calculate these values.

```r
median(happy_numeric$love)
```

```
## [1] 3
```

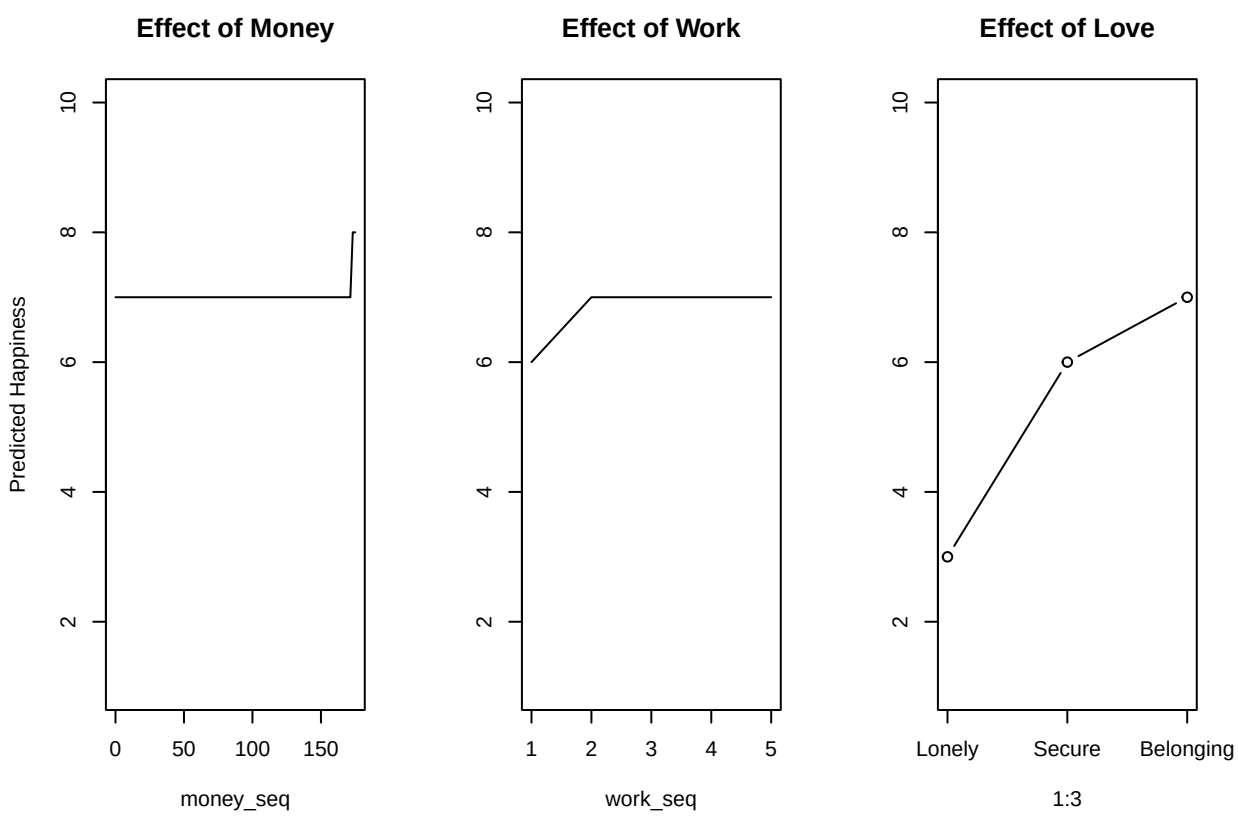
```r
summary(happy)
```

```
##      happy            money                     sex            love
##  Min.   : 2.000   Min.   :  0.00   Unsatisfactory:12   Lonely   : 3
##  1st Qu.: 5.000   1st Qu.: 42.50   Satisfactory  :27   Secure   :15
##  Median : 7.000   Median : 50.00                       Belonging:21
##  Mean   : 6.744   Mean   : 62.15
##  3rd Qu.: 8.000   3rd Qu.: 78.00
##  Max.   :10.000   Max.   :175.00
##       work
##  Min.   :1.000
##  1st Qu.:3.000
##  Median :4.000
##  Mean   :3.359
##  3rd Qu.:4.000
##  Max.   :5.000
```

Median for money is 50. Median for sex is satisfactory. Median of love is level 3, belonging. Median of work is 4.

```r
median_money <- 50
median_sex <- "Satisfactory"
median_love <- "Belonging"
median_work <- 4

# SCENARIO A: VARYING MONEY
money_seq <- seq(min(happy$money), max(happy$money), length.out = 100)
df_money <- data.frame(money = money_seq, sex = median_sex, love = median_love, work = median_work)
```

```r
prob_money <- predict(model_prop_odds, newdata = df_money, type = "probs")
# Get most likely category (the column index with max prob)
mode_money <- apply(prob_money, 1, which.max)

# SCENARIO B: VARYING WORK
work_seq <- seq(min(happy$work), max(happy$work), length.out = 5)
df_work <- data.frame(money = median_money, sex = median_sex, love = median_love, work = work_seq)
prob_work <- predict(model_prop_odds, newdata = df_work, type = "probs")
mode_work <- apply(prob_work, 1, which.max)

# SCENARIO C: VARYING LOVE
love_levels <- c("Lonely", "Secure", "Belonging")
df_love <- data.frame(money = median_money, sex = median_sex, love = love_levels, work = median_work)
prob_love <- predict(model_prop_odds, newdata = df_love, type = "probs")
mode_love <- apply(prob_love, 1, which.max)

# Visual Comparison
par(mfrow=c(1,3))
plot(money_seq, mode_money, type="l", main="Effect of Money", ylim=c(1,10), ylab="Predicted Happiness")
plot(work_seq, mode_work, type="l", main="Effect of Work", ylim=c(1,10), ylab="")
plot(1:3, mode_love, type="b", main="Effect of Love", ylim=c(1,10), xaxt="n", ylab="")
axis(1, at=1:3, labels=c("Lonely", "Secure", "Belonging"))
```



We see not much effect of money, which makes sense since the coefficient is not large. Work is also just a little. The most is love. This matches our findings earlier.

g.)

Predict the happiness distribution for subject whose parents earn $30,000 a year, who is lonely, not sexually active and has no job.

```
our_subject <- data.frame(
  money = 30, # 30 thousand dollars
  love = "Lonely", # lonely
  sex = "Unsatisfactory", # no sex
  work = 1 # no job
)

pred_our_subject <- predict(model_prop_odds, newdata = our_subject, type = "probs")
print(pred_our_subject)
```

```
##            2            3            4            5            6            7
## 5.046777e-01 2.245143e-01 2.388386e-01 2.635228e-02 2.352246e-03 2.876042e-03
##            8            9           10
## 3.804580e-04 6.874443e-06 1.517076e-06
```

```
# Visualize the distribution
barplot(pred_our_subject, main = "Predicted Happiness Distribution",
        xlab = "Happiness Level (1-10)", ylab = "Probability",
        col = "tomato", border = "white")
```

They likely have 2 happiness.

Problem 2

As part of a housing market study, you collected data on the number of apartments sold at various price ranges (low, medium, high), in different locations (District A, B, C and D), of different building types (high-rise, low-rise, townhouse), and for three buyer incomes (low, medium, high). You'd like to model the probability of selling an apartment in a specific price range, as function of the remaining three characteristics. You consider two models: (1) price range is determined by additive contributions of location, buyer income, and their interaction and (2) price range is independent of all other factors.

a.)

You only have access to software which fits Poisson regressions. Write models (1) and (2) above in terms of the Poisson log-linear model. Clearly define the notation and state the assumptions.

This is a surrogate log-linear model that we model with glm poisson. The model tests that price is independent of all other factors, and is the same as the multinomial null model (ie like a contingency table).

The model predictors:

Let n_{ijkl} be the observed count (number of apartments sold), where the indices represent:i: - Price Range ($i = 1, \ldots, I$) The Response Variable - j: Location ($j = 1, \ldots, J$) - k: Building Type ($k = 1, \ldots, K$) - l: Buyer Income ($l = 1, \ldots, L$)

We assume each count n_{ijkl} follows a Poisson distribution with mean μ_{ijkl}. In a logit log-linear model, we model $\log(\mu_{ijkl})$. We also assume log-linearity in that the log of the expected count is a linear function of the parameters. There is multinomial equivalence, ie that there is a fixed amount that is split into the categories.

Model 1

For model 1 with conditional independence, we have to model each predictor used:

$$\log(\mu_{ijkl}) = \beta_0 + \beta_i^P + \beta_j^L + \beta_k^B + \beta_l^I + \beta_{jk}^{LB} + \beta_{jl}^{LI} + \beta_{kl}^{BI} + \beta_{jkl}^{LBI} + \beta_{\mathbf{ij}}^{\mathbf{PL}} + \beta_{\mathbf{il}}^{\mathbf{PI}} + \beta_{\mathbf{ijl}}^{\mathbf{PLI}}$$

We care about how the other predictors (location, buyer income) and price have interaction. We do not care about building type, but we can include its interactions if we wish to test its interaction with price.

Model 2

For model 2 of global independence, we assume the apartment price range i does not change based on the predictors we have.

$$\log(\mu_{ijkl}) = \beta_0 + \beta_i^P + \beta_{jkl}^{LBI}$$

We have predictor for main effect of price and the join distribution of all other explanatory variables. There is no interaction between other variables and price.

b.)

> What test statistic would you use to assess the fit of each model? Make sure to state the test statistic's reference distribution (including degrees of freedom).

Both models are poisson glm, so we can use deviance or pearson's residual chi-squared. Under the null hypothesis, deviance and Pearson's chi-squared follow a Chi-squared distribution.

The degrees of freedom is calculated as following: $N = I \times J \times K \times L = 108$ and $p =$ the number of non-redundant parameters in your model (including the intercept λ). Then $df = N - p$.

For model 1, we have parameters: $J \times K \times L = 4 \times 3 \times 3 = \mathbf{36}$, price main effect: $(I-1) = 2$, price $\times$ location: $(I-1)(J-1) = 2 \times 3 = 6$Price $\times$, buyer income: $(I-1)(L-1) = 2 \times 2 = 4$, price $\times$ location $\times$, buyer income: $(I-1)(J-1)(L-1) = 2 \times 3 \times 2 = 12$. So, $df = 108 - 60 = 48$.

For model 2, we have intercept (λ): 1, price effect (λ_i^P): $I - 1 = 2$, joint explanatory effect (λ_{jkl}^{LBI}): $(J \times K \times L) - 1 = (4 \times 3 \times 3) - 1 = 35$. We find $df = 108 - 38 = 70$.

Problem 3

A computer software company classifies a random sample of their service calls based on the complexity of the issue (1=low to 4=high) and the customer's satisfaction with the solution (1=very low to 5=very high).

a.)

Are these categorical variables nominal or ordinal? Explain your answer.

Complexity of issue is ordinal. It is ranked from low to high, ie 1 is smaller than 2 etc.

Customer satisfaction is ordinal. They also go from very low to very high, which means they follow this ranked hierarchy again.

b.)

The company would like to develop a model to predict satisfaction based on complexity. Fit the following four models and use the results to determine the best model. Make sure to summarize all tests used in this determination.

i.) Proportional-odds model with complexity as a nominal factor

```r
# our data
df <- data.frame(
  Complexity = rep(1:4, each=5),
  Satisfaction = rep(1:5, 4),
  Freq = c(22, 64, 88, 47, 9, 32, 91, 170, 116, 41, 26, 90, 242, 322, 110, 21, 78, 248, 585, 176)
)
print(df)
```

```
##    Complexity Satisfaction Freq
## 1           1            1   22
## 2           1            2   64
## 3           1            3   88
## 4           1            4   47
## 5           1            5    9
## 6           2            1   32
## 7           2            2   91
## 8           2            3  170
## 9           2            4  116
## 10          2            5   41
## 11          3            1   26
## 12          3            2   90
## 13          3            3  242
## 14          3            4  322
## 15          3            5  110
## 16          4            1   21
## 17          4            2   78
## 18          4            3  248
```

```
## 19          4            4  585
## 20          4            5  176
```

Again, we use polr from MASS and now multinomial from nnet libraries like Professor Song.

In models i and ii, we treat complexity as a ordered 4 levels. In models iii and iv, we ignore ordering and treat them as categories that the counts fall into.

```r
library(MASS) # For polr
library(nnet) # For multinomial

df$Satisfaction <- factor(df$Satisfaction, ordered = TRUE)

# i) Proportional-odds with complexity as nominal
mod_i <- polr(Satisfaction ~ factor(Complexity), weights = Freq, data = df, Hess = TRUE)
summary(mod_i)
```

```
## Call:
## polr(formula = Satisfaction ~ factor(Complexity), data = df,
##     weights = Freq, Hess = TRUE)
##
## Coefficients:
##                      Value Std. Error t value
## factor(Complexity)2 0.500     0.1469   3.404
## factor(Complexity)3 1.300     0.1378   9.433
## factor(Complexity)4 1.769     0.1348  13.123
##
## Intercepts:
##     Value    Std. Error t value
## 1|2  -2.1361   0.1448   -14.7538
## 2|3  -0.5001   0.1206    -4.1465
## 3|4   1.0723   0.1230     8.7166
## 4|5   3.2717   0.1354    24.1692
##
## Residual Deviance: 6839.218
## AIC: 6853.218
```

```r
# ii) Proportional-odds with complexity as ordered
mod_ii <- polr(Satisfaction ~ as.numeric(Complexity), weights = Freq, data = df, Hess = TRUE)
summary(mod_ii)
```

```
## Call:
## polr(formula = Satisfaction ~ as.numeric(Complexity), data = df,
##     weights = Freq, Hess = TRUE)
##
## Coefficients:
##                          Value Std. Error t value
## as.numeric(Complexity) 0.6038    0.03844   15.71
##
## Intercepts:
##     Value    Std. Error t value
## 1|2  -1.5188   0.1439   -10.5517
## 2|3   0.1148   0.1199     0.9577
```

```
## 3|4   1.6834   0.1251    13.4510
## 4|5   3.8842   0.1419    27.3732
##
## Residual Deviance: 6843.134
## AIC: 6853.134
```

```r
# iii) Multinomial logit with complexity as nominal
mod_iii <- multinom(Satisfaction ~ factor(Complexity), weights = Freq, data = df)
```

```
## # weights:  25 (16 variable)
## initial  value 4149.130938
## iter  10 value 3464.883824
## iter  20 value 3408.234938
## final  value 3408.230314
## converged
```

```r
summary(mod_iii)
```

```
## Call:
## multinom(formula = Satisfaction ~ factor(Complexity), data = df,
##     weights = Freq)
##
## Coefficients:
##   (Intercept) factor(Complexity)2 factor(Complexity)3 factor(Complexity)4
## 2   1.0678513          -0.0227385           0.1739006           0.2444184
## 3   1.3862735           0.2837945           0.8446130           1.0827239
## 4   0.7591081           0.5287197           1.7573892           2.5680639
## 5  -0.8938743           1.1416998           2.3363038           3.0199293
##
## Std. Errors:
##   (Intercept) factor(Complexity)2 factor(Complexity)3 factor(Complexity)4
## 2   0.2471416           0.3214308           0.3326459           0.3486009
## 3   0.2383654           0.3065128           0.3152989           0.3293529
## 4   0.2583227           0.3264985           0.3290887           0.3406804
## 5   0.3956904           0.4606636           0.4518019           0.4581220
##
## Residual Deviance: 6816.461
## AIC: 6848.461
```

```r
# iv) Multinomial logit with complexity as ordered
mod_iv <- multinom(Satisfaction ~ as.numeric(Complexity), weights = Freq, data = df)
```

```
## # weights:  15 (8 variable)
## initial  value 4149.130938
## iter  10 value 3434.506133
## final  value 3411.150086
## converged
```

```r
summary(mod_iv)
```

```
## Call:
```

```
## multinom(formula = Satisfaction ~ as.numeric(Complexity), data = df,
##     weights = Freq)
##
## Coefficients:
##   (Intercept) as.numeric(Complexity)
## 2   0.9176277              0.0976094
## 3   0.9833803              0.3823607
## 4  -0.4009903              0.9398080
## 5  -1.5482994              0.9365307
##
## Std. Errors:
##   (Intercept) as.numeric(Complexity)
## 2   0.2921758              0.1084028
## 3   0.2763538              0.1015624
## 4   0.2882384              0.1032467
## 5   0.3438475              0.1166823
##
## Residual Deviance: 6822.3
## AIC: 6838.3
```

AIC of model iv is lowest while model 3 has lowest residual deviance.

We can perform more robust comparison test using residual deviances in the likelihood ratio test.

For model 1 vs 2: $G^2 = 6843.134 - 6839.218 \approx 3.9$ with $df = 7 - 5 = 2$ and p-value of 0.1422741. We do not reject the null hypothesis and accept that treating complexity as numeric is sufficient. We can also use ANOVA to do this likelihood ratio test.

```r
anova(mod_i, mod_ii)
```

```
## Likelihood ratio tests of ordinal regression models
##
## Response: Satisfaction
##                     Model Resid. df Resid. Dev   Test    Df LR stat.  Pr(Chi)
## 1 as.numeric(Complexity)      2573   6843.134
## 2     factor(Complexity)      2571   6839.218 1 vs 2     2 3.915553 0.141172
```

For model 1 vs 3: $G^2 = 6843.134 - 6816.461 \approx 26.7$ and $df = 16 - 7 = 9$ with p-value of 0.001567796. Model 3 is significantly better and this likely suggests that the proportional odds assumption model should be rejected and we should use the multinomial logit models instead.

Finally, 3 vs 4:

```r
anova(mod_iii, mod_iv)
```

```
## Likelihood ratio tests of Multinomial Models
##
## Response: Satisfaction
##                     Model Resid. df Resid. Dev   Test    Df LR stat.   Pr(Chi)
## 1 as.numeric(Complexity)        72   6822.300
## 2     factor(Complexity)        64   6816.461 1 vs 2     8 5.839544 0.6651998
```

4 is not statistically significantly better, but it uses less parameters, leading to a better degree of freedom and a simpler model that also performs the best (with lowest AIC).

We choose model 4.

c.)

> The residual deviance using both polr and multinom procedures provide the ungrouped residual deviance, which cannot be used to assess goodness of fit. What is the grouped residual deviance and degrees of freedom for your best fitting model in b)?

We have to calculate this grouped residual deviance. First for the fully saturated model deviance calculation:

```
call_counts = matrix(c(22, 64, 88, 47, 9, 32, 91, 170, 116, 41, 26, 90, 242, 322, 110, 21, 78, 248, 585
                       nrow = 4, byrow = TRUE)
row_totals <- rowSums(call_counts)
proportions <- call_counts / row_totals
deviance_log_likelihood_saturated <- 2*sum(call_counts * log(proportions))
df_saturated <- 4 * (5-1)

deviance_log_likelihood_modeliv <- 2*logLik(mod_iv)

grouped_deviance <- deviance_log_likelihood_saturated - deviance_log_likelihood_modeliv
grouped_df <- df_saturated - length(coef(mod_iv)) # should be 16 - 8 = 8

pchisq(grouped_deviance, grouped_df, lower.tail = FALSE)
```

```
## 'log Lik.' 0.6651998 (df=8)
```

Since the p-value is greater than 0.05, we fail to reject the null hypothesis, which means that we accept the model as providing a good fit to our data.

d.)

> Given your best model in b), compute the conditional distribution of satisfaction given the complexity of the issue is high.

We predict on complexity with level high:

```
high_complexity_data <- data.frame(Complexity = 4)

predict(mod_iv, newdata = high_complexity_data, type = "probs")
```

```
##          1          2          3          4          5
## 0.01825393 0.06752147 0.22524827 0.52459257 0.16438376
```

We are most likely going to be in satisfaction level 4.

Problem 4

The following data report the survival of 30 late stage liver cancer patients (in weeks) according to three different treatment methods. Some R output is provided below.

a.)

Compute the Kaplan-Meier estimates of the survival function for each of three treatments.

From our notes, we know:

$$S(t) = \prod_{i:t_i \leq t} \left(1 - \frac{d_i}{n_i}\right)$$

We can do these calculation for each treatment.

Treatement 1:

Time (t)	At Risk (n_i)	Events (d_i)	Censored	$1 - (d_i/n_i)$	$\hat{S}(t)$
0	10	0	0	1	1.000
2	10	1	1	0.900	0.900
4	8	1	0	0.875	0.788
7	6	1	0	0.833	0.656
9	5	1	0	0.800	0.525
12	4	1	0	0.750	0.394
24	1	1	0	0.000	0.000

Treatment 2:

Time (t)	At Risk (n_i)	Events (d_i)	Censored	$1 - (d_i/n_i)$	$\hat{S}(t)$
0	10	0	0	1	1.000
1	10	2	0	0.800	0.800
3	8	1	0	0.875	0.700
6	6	2	0	0.667	0.467
9	3	1	0	0.667	0.311
10	2	1	0	0.500	0.156

Treatment 3:

Time (t)	At Risk (n_i)	Events (d_i)	Censored	$1 - (d_i/n_i)$	$\hat{S}(t)$
0	10	0	0	1	1.000
6	9	1	0	0.889	0.889
8	8	1	0	0.875	0.778
10	7	1	0	0.857	0.667
13	5	2	0	0.600	0.400
19	2	1	0	0.500	0.200
21	1	1	0	0.000	0.000

Alternatively, we can use library survival and write our data and test it using survfit function. Honestly easier to just use a excel sheet and convert to latex like we did here.

```r
library(survival)
```

```
##
## Attaching package: 'survival'

## The following objects are masked from 'package:faraway':
##
##     rats, solder
```

```r
library(survminer)
```

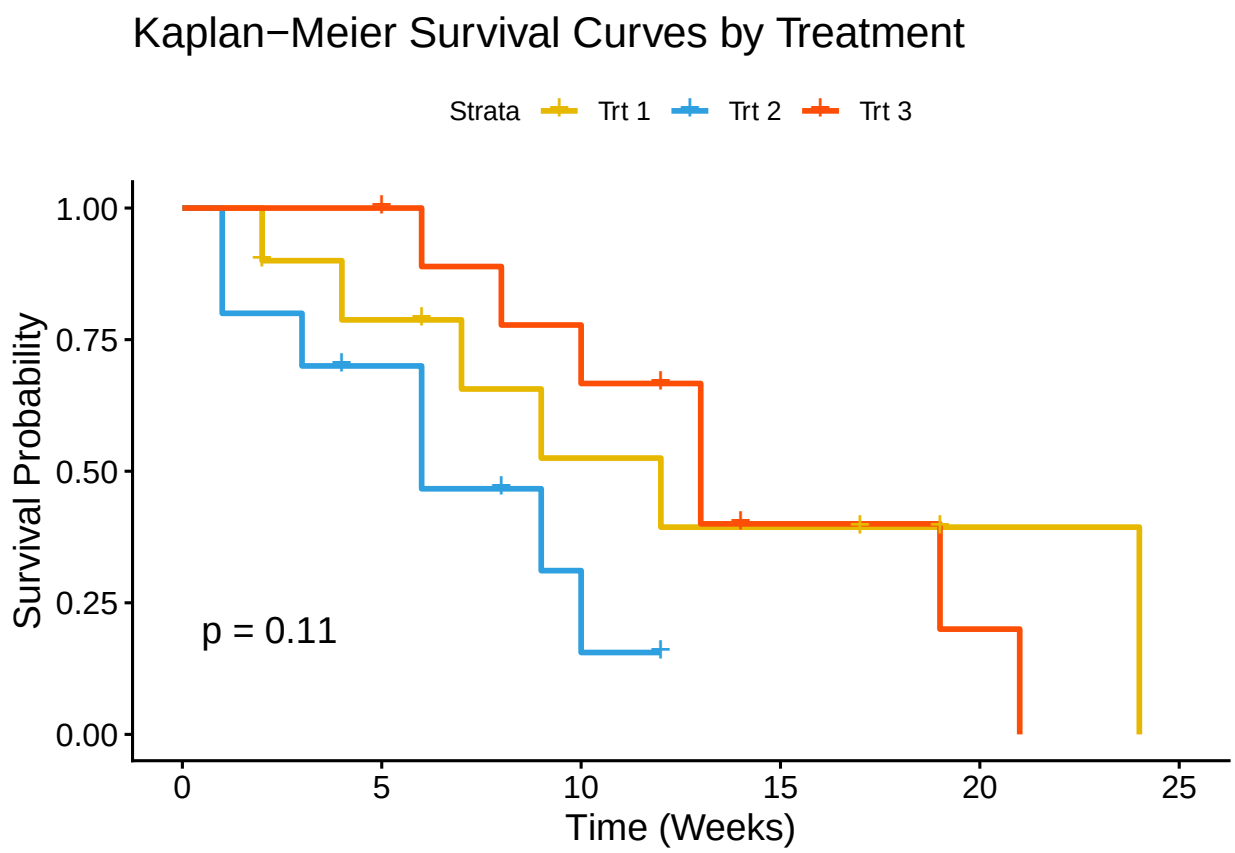
```
## Warning: package 'survminer' was built under R version 4.5.3

## Loading required package: ggplot2

## Loading required package: ggpubr

## Warning: package 'ggpubr' was built under R version 4.5.3

##
## Attaching package: 'survminer'

## The following object is masked from 'package:survival':
##
##     myeloma
```

```r
# status: 1 = death (event), 0 = censored
trt1_time <- c(2, 2, 4, 6, 7, 9, 12, 17, 19, 24)
trt1_cens <- c(0, 1, 1, 0, 1, 1, 1, 0, 0, 1)

trt2_time <- c(1, 1, 3, 4, 6, 6, 8, 9, 10, 12)
trt2_cens <- c(1, 1, 1, 0, 1, 1, 0, 1, 1, 0)

trt3_time <- c(5, 6, 8, 10, 12, 13, 13, 14, 19, 21)
trt3_cens <- c(0, 1, 1, 1, 0, 1, 1, 0, 1, 1)

df <- data.frame(
  time = c(trt1_time, trt2_time, trt3_time),
  status = c(trt1_cens, trt2_cens, trt3_cens),
  trt = rep(c("Trt 1", "Trt 2", "Trt 3"), each = 10)
)

# Compute KM estimate
km_fit <- survfit(Surv(time, status) ~ trt, data = df)
print(summary(km_fit))
```

```
## Call: survfit(formula = Surv(time, status) ~ trt, data = df)
##
##                 trt=Trt 1
##  time n.risk n.event survival std.err lower 95% CI upper 95% CI
```

```
##     2     10       1    0.900  0.0949        0.732        1.000
##     4      8       1    0.787  0.1340        0.564        1.000
##     7      6       1    0.656  0.1638        0.402        1.000
##     9      5       1    0.525  0.1759        0.272        1.000
##    12      4       1    0.394  0.1742        0.165        0.937
##    24      1       1    0.000     NaN           NA           NA
##
##                 trt=Trt 2
##  time n.risk n.event survival std.err lower 95% CI upper 95% CI
##     1     10       2    0.800   0.126       0.5868        1.000
##     3      8       1    0.700   0.145       0.4665        1.000
##     6      6       2    0.467   0.166       0.2326        0.936
##     9      3       1    0.311   0.168       0.1077        0.899
##    10      2       1    0.156   0.139       0.0272        0.891
##
##                 trt=Trt 3
##  time n.risk n.event survival std.err lower 95% CI upper 95% CI
##     6      9       1    0.889   0.105       0.7056        1.000
##     8      8       1    0.778   0.139       0.5485        1.000
##    10      7       1    0.667   0.157       0.4200        1.000
##    13      5       2    0.400   0.174       0.1707        0.938
##    19      2       1    0.200   0.166       0.0393        1.000
##    21      1       1    0.000     NaN           NA           NA
```

```r
# Plot with some online help
ggsurvplot(km_fit,
           data = df,
           pval = TRUE,              # Adds Log-Rank p-value to the plot
           conf.int = FALSE,
           palette = c("#E7B800", "#2E9FDF", "#FC4E07"),
           title = "Kaplan-Meier Survival Curves by Treatment",
           xlab = "Time (Weeks)",
           ylab = "Survival Probability",
           legend.labs = c("Trt 1", "Trt 2", "Trt 3"))
```



b.)

Test whether the Kaplan-Meier estimates of the three survival functions are equal.

Log-Rank test (in R) for Equality

```
surv_test <- survdiff(Surv(time, status) ~ trt, data = df)
print(surv_test)
```

```
## Call:
## survdiff(formula = Surv(time, status) ~ trt, data = df)
##
##             N Observed Expected (O-E)^2/E (O-E)^2/V
## trt=Trt 1 10        6     7.63     0.348     0.656
## trt=Trt 2 10        7     3.61     3.171     4.482
## trt=Trt 3 10        7     8.76     0.352     0.708
##
##  Chisq= 4.5  on 2 degrees of freedom, p= 0.1
```

High p value greater than 0.05 suggests that we do not reject the null hypothesis. The 3 treatments are statistically not significantly different.

c.)

> Assume the survival functions are "exponential." Compute the maximum likelihood estimates of the mean survival time for the three groups as well as their standard errors, respectively.

We assume a constant hazard rate. Given $S(t) = e^{-\lambda t}$, mean survival time is given by $\theta = 1/\lambda$. We simply use the MLE estimate for the mean here $\hat{\theta} = \frac{\sum t_i}{d}$ with corresponding Standard Error estimate of $SE(\hat{\theta}) = \frac{\hat{\theta}}{\sqrt{d}}$. d is the count of how many die (ie total minus the censored).

Group 1:

```
sum(trt1_time)
```

```
## [1] 102
```

```
sum(trt1_cens)
```

```
## [1] 6
```

```
sum(trt1_time)/sum(trt1_cens)
```

```
## [1] 17
```

```
sum(trt1_time)/sum(trt1_cens) / sqrt(sum(trt1_cens))
```

```
## [1] 6.940221
```

MLE Mean is 17 weeks and SE is 6.94.

Group 2:

```
sum(trt2_time)
```

```
## [1] 60
```

```
sum(trt2_cens)
```

```
## [1] 7
```

```
sum(trt2_time)/sum(trt2_cens)
```

```
## [1] 8.571429
```

```
sum(trt2_time)/sum(trt2_cens) / sqrt(sum(trt2_cens))
```

```
## [1] 3.239695
```

MLE Mean is 8.571429 weeks and SE is 3.239695.

Group 3:

```r
sum(trt3_time)
```

```
## [1] 121
```

```r
sum(trt3_cens)
```

```
## [1] 7
```

```r
sum(trt3_time)/sum(trt3_cens)
```

```
## [1] 17.28571
```

```r
sum(trt3_time)/sum(trt3_cens) / sqrt(sum(trt3_cens))
```

```
## [1] 6.533386
```

MLE Mean is 17.28571 weeks and SE is 6.533386

d.)

> Test whether the mean survival time for the three groups are equal.

We test likelihood ratio test for exponential models. Compare different mean survival vs all the same models.

If they all have the same mean, we have combined model of total events $(D) = 6+7+6 = 19$ and total time $(T) = 102 + 60 + 121 = 283$ which gives us $\hat{\theta}_{common} = 283/19 \approx 14.89$. The log likelihood of this is $L_0) = D\ln(1/\hat{\theta}_{common}) - \frac{T}{\hat{\theta}_{common}} = 19\ln(1/14.89) - 19 \approx -70.31$.

For all different means:

$$L_1 = \sum[d_j \ln(1/\hat{\theta}_j) - \frac{T_j}{\hat{\theta}_j}] = \sum[d_j \ln(1/\hat{\theta}_j) - d_j] = [6\ln(1/17) - 6] + [7\ln(1/8.57) - 7] + [6\ln(1/20.17) - 6] \approx -69.06$$

Our test statistic is $G^2 = -2(-70.31 - (-69.06)) = 2.5$ and $df = 3 - 1 = 2$. This has p-value greater than 0.05.

Compare to R results:

```r
# Fit the null model (common mean)
fit_null <- survreg(Surv(time, status) ~ 1, data = df, dist = "exponential")

# Fit the alternative model (different means)
fit_alt <- survreg(Surv(time, status) ~ trt, data = df, dist = "exponential")

# Likelihood Ratio Test
anova(fit_null, fit_alt)
```

```
##   Terms Resid. Df    -2*LL Test Df Deviance  Pr(>Chi)
## 1     1        29 145.9886      NA       NA        NA
## 2   trt        27 143.9750       2 2.013617 0.3653832
```

We fail to reject the null hypothesis. There is statistically no significant difference between mean survival times of the three groups.